

Question number	Answer	Additional guidance	Mark
7(a)	(carbon dioxide) burning (fossil) fuels / farming animals / burning trees / burning plastics / vehicle emissions and farming rice / paddy fields / farming ruminants / farming named ruminant e.g. cattle / landfill sites / fossil fuel production / coal mining / burning biomass / burning biofuels / burning plastics / mismanagement of sewage / draining of peat bogs (1)	NB answers must be different for the mark to be awarded ACCEPT burning {coal / oil / natural gas / peat} ACCEPT livestock for farming	(1)

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7(b)	A description that includes three of the following points: - greenhouse gases {accumulate / increase / more released / in excess} in the (upper) atmosphere (1) - allow {UV / short wavelength / visible} light {through / to reach Earth's surface} (1) - but {prevent the exit of / trap / absorb} {IR (radiation) / long wavelength light / heat (energy)} (1) - causing the Earth's {atmosphere / surface} to {heat up / increase in temperature} (1)	ACCEPT named greenhouse gas ACCEPT reflect (back) / re-emit	(3)

Question number	Answer	Additional guidance	Mark
7(c)	value between 1.10 and 1.40 up to 2 decimal places (1)		(1)

Question number	Answer	Additional guidance	Mark
7(d)(i)	9.5 (1)		(1)

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7(d)(ii)	An answer that includes three of the following points: - tackling the release of nitrous oxide is urgent as it has {a (very) high / the highest} WP (1) - the issue with carbon dioxide production is it has a long-term effect (1) - reducing methane because as it has a (fairly) high WP (1) - tackle nitrous oxide first, carbon dioxide second, methane third / all gases should be tackled as they all affect climate change (1)	ACCEPT carbon dioxide would be a more long-term solution as the {WP is low / it remains in the atmosphere for years} ACCEPT methane has least effect as intermediate WP and lasts 12 years	(3)

Question number	Answer	Additional guidance	Mark
7(e)(i)	An explanation that includes two of the following points: - the higher the turbidity the more {cells / bacteria / growth of bacteria} present (1) - use lines of best fit (1) - compare the gradients of the two lines (1) OR - the higher the turbidity the more {cells / bacteria / growth of bacteria} present (1) - compare the mean rate of change in turbidity (1) - using {the values at 1 and 14 days / $y = m x + c$} (1)	IGNORE doubling time / GRC ACCEPT lines of best fit shown on the graph ACCEPT B has steeper gradient (so higher growth rate)	(2)

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7(e)(ii)	{energy / carbon} is not enough to sustain the metabolic requirements of the bacteria OR - concentration outside cell is too low for (facilitated) diffusion of methane (into the cell) OR - bacteria are not able to actively transport methane (into the cell) (1)	ACCEPT named use e.g. cell division, protein synthesis, reproduction IGNORE growth / biomass	(1)

Question number	Answer	Additional guidance	Mark
7(e)(iii)	An explanation that includes the following points: - because {nitrous oxide / carbon dioxide / other greenhouse gases (than methane) / greenhouse gases} may be produced (by the bacteria) (1) {nitrous oxide / carbon dioxide} could cause more {climate change / global warming} than the reduction in methane would reduce (1)	ACCEPT more greenhouse gases maybe produced than (methane) absorbed {nitrous oxide / carbon dioxide} contribute to acid rain	(2)

